

Go to:

<https://cearun.grc.nasa.gov/cgi-bin/CEARUN/setProblemType.cgi>

ROCKET PROBLEM (RO)

Problem Type | Pressures | Fuel(s) | Oxidizer(s) | Oxid/Fuel | Exit Conditions | Final

Fuel(s): Pure H2
Oxid(s): Pure O2

Set Type of Problem for CEA to perform: Chemical Equilibrium Problem Types

Type Code	Description
☒ rocket	Rocket
☐ hp	Assigned Enthalpy & Pressure
☐ tp	Assigned Temperature & Pressure
☐ det	Chapman-Jouguet Detonation
☐ shock	Shock Tube
☐ tv	Assigned Temperature & Density
☐ uv	Combustion at Assigned Density
☐ sp	Assigned Entropy & Pressure
☐ sv	Assigned Entropy & Density

Change Problem Type & Continue to Next Form.

ROCKET PROBLEM (RO)

Problem Type | Pressures | Fuel(s) | Oxidizer(s) | Oxid/Fuel | Exit Conditions | Final

Fuel(s): Pure H2
Oxid(s): Pure O2

Enter Pressure Values

Use either Low-High-Interval fields (on left side) or numbered fields (on right side)—*Not both.*

Pressure

Enter Low/High/Interval values for no more than 24 datapoints.

Low Value: 100

High Value: 210

Interval: 55

Clear Low/High/Int. Fields

1.		9.		17.	
2.		10.		18.	
3.		11.		19.	
4.		12.		20.	
5.		13.		21.	
6.		14.		22.	
7.		15.		23.	
8.		16.		24.	

Clear Numbered Fields

Select one: ☒ atm ☐ bar ☐ mmHg ☐ psia

*Ref: [NASA RPI311 Part II \(Users Manual\)](#), p. 13, Section 2.4.8

ROCKET PROBLEM (R0)

Problem Type | Pressures | Fuel(s) | Oxidizer(s) | Oxid/Fuel | Exit Conditions | Final

Fuel(s): Pure H2
 Oxid(s): Pure O2

Select Your Fuel(s)

Select one of the following:

Select one of the following compounds for *simple* (1-component) fuels, or select a mixture using the periodic table:

☐ CH4
 ☐ CH4(L)
 ☒ H2
 ☐ H2(L)
 ☐ RP-1
 ☐ Use Periodic Table (mixtures)

- The species listed above are assumed to be **pure**. If your reactant is not shown here, or you need to blend one or more compounds, use the Periodic Table.
- Please note that any fuel combinations using the Periodic Table will cancel out a current simple-fuel selection.**
- Be careful to select the appropriate compounds. Some compounds are represented in the CEA database in more than one form. For example, H2 refers to gas while H2(L) is liquid.
- To specify reactants without distinguishing between 'fuels' and 'oxidants', select '**None**' from the Oxidizer Selection page, and CEA will be instructed to skip the Oxidizer Selection Form.

Enter Reactant Temperature(K), if needed:

Please specify how to define reactant mixtures (*both Fuels & Oxidizers*): ☒ wt% ☐ mole

Accept fuel Selection & Continue Reset

ROCKET PROBLEM (R0)

Problem Type | Pressures | Fuel(s) | Oxidizer(s) | Oxid/Fuel | Exit Conditions | Final

Fuel(s): Pure H2
 Oxid(s): Pure O2

Select Your Oxidizer(s)

Select one of the following:

Select one of the following compounds for *simple* (1-component) oxidizers, or select a mixture using the periodic table:

☐ Air
 ☐ CL2
 ☐ CL2(L)
 ☐ F2
 ☐ F2(L)
 ☐ H2O2(L)
 ☐ N2H4(L)
 ☐ N2O
 ☐ NH4NO3(I)
 ☒ O2
 ☐ O2(L)

Enter Oxidizer Temperature(K), if needed:

☐ Use Periodic Table (mixtures)
 ☐ None

- Use the Periodic Table if your oxidizer does not appear above or you need a mixture of two or more compounds.
- Please note that selecting oxidizer(s) using the Periodic Table cancels out any simple-oxidizer selection.**
- If you do not want your analysis to distinguish between fuels and oxidizers, select **None**.
- Select your reactants carefully, since some compounds are represented in the CEA Database in more than one form. For example, H2O refers to water vapor while H2O(L) is liquid.

Accept Oxidizer Selection & Continue to next form. Reset

ROCKET PROBLEM (R0)

Problem Type | Pressures | Fuel(s) | Oxidizer(s) | Oxid/Fuel | Exit Conditions | Final

Fuel(s): Pure H2
 Oxid(s): Pure O2

Select Your Oxidizer(s)

Select one of the following:

Select one of the following compounds for *simple* (1-component) oxidizers, or select a mixture using the periodic table:

☐ Air
 ☐ CL2
 ☐ CL2(L)
 ☐ F2
 ☐ F2(L)
 ☐ H2O2(L)
 ☐ N2H4(L)
 ☐ N2O
 ☐ NH4NO3(I)
 ☒ O2
 ☐ O2(L)

Enter Oxidizer Temperature(K), if needed:

☐ Use Periodic Table (mixtures)
 ☐ None

- Use the Periodic Table if your oxidizer does not appear above or you need a mixture of two or more compounds.
- Please note that selecting oxidizer(s) using the Periodic Table cancels out any simple-oxidizer selection.**
- If you do not want your analysis to distinguish between fuels and oxidizers, select **None**.
- Select your reactants carefully, since some compounds are represented in the CEA Database in more than one form. For example, H2O refers to water vapor while H2O(L) is liquid.

Accept Oxidizer Selection & Continue to next form. Reset

ROCKET PROBLEM (RO)

Problem TypePressuresFuel(s)Oxidizer(s)Oxid/FuelExit ConditionsFinal

Fuel(s): Pure H2
Oxid(s): Pure O2

Enter Proportions of Oxidizer/Fuel (Optional)

How do you want to specify the Oxidizer/Fuel Ratio?

☐ %fuel: %Fuel by Weight

☒ o/f: Oxidizer/Fuel Wt. ratio

☐ phi: Equivalence based on Fuel/Oxid. wt ratio (Eq 9.19*)

☐ r,eq.ratio: Equivalence based on Valence (Eq 9.18*)

If these values are not filled, CEA will assume 1:1 Oxidizer/Fuel ratio by weight.
*Reference: [NASA RP1311 Part II \(Users Manual\), p. 13, Sect. 2.4.3](#)

Enter Low-Value, High-Value and Interval

Low Value: 1

High Value: 8

Interval: 1

Clear Low/High/Int. Fields

The Low-High-Interval values must result in an array of no more than 30 datapoints.

Or, Fill in these numbered fields. Do not use both sides.

1.

2.

3.

4.

5.

6.

7.

8.

9.

10.

11.

12.

13.

14.

15.

16.

17.

18.

19.

20.

21.

22.

23.

24.

25.

26.

27.

28.

29.

30.

Clear Numbered Fields

Accept Input & Continue to Next Form. Reset

ROCKET PROBLEM (RO)

Problem TypePressuresFuel(s)Oxidizer(s)Oxid/FuelExit ConditionsFinal

Fuel(s): Pure H2
Oxid(s): Pure O2

Define Exit Conditions (Optional)

Pc/Pe (Chamber/Exit) Pressure Ratios

Values will be sorted in Ascending order.

1.

2.

3.

4.

5.

6.

7.

8.

9.

10.

11.

12.

13.

14.

15.

16.

Clear Numbered Fields

Subsonic Area Ratios

Values will be sorted in Descending order.

1.

2.

3.

4.

5.

6.

7.

8.

9.

10.

11.

12.

13.

Clear Numbered Fields

Supersonic Area Ratios

Values will be sorted in Ascending order.

1.

2.

3.

4.

5.

6.

7.

8.

9.

10.

11.

12.

13.

Clear Numbered Fields

Accept Input & Continue to Next Form. Reset

Enter Your Final Choices Before Running CEA.

Select Your Output File Length:

- ☒ Short: Prints only error messages and final tables, excluding atom ratios and species being considered during calculation.
☐ Long: Prints all output tables.
☐ Debug: Includes intermediate output.

CEA Options

Express **Products** as: ☐ Mass-Fractions ☒ Mole-Fractions

Express heat as: ☒ SI units ☐ Calories

☐ Include Transport Properties?

Ref: [NASA RP1311 Part I \(Analysis\), Ch. 5](#)

☐ Consider Ionized Species as possible products?

☐ Set Trace Value: x 10[^]-

The Trace Option prints species compositions having mole or mass fractions exceeding the Trace Value. The criteria for equilibrium composition convergence will be tighter to ensure accuracy. Mass- and mole-fractions will be expressed in E-format (eg, '2.0089-4' = '0.0002').

Rocket Problem Options:

Select one or both: ☒ Equilibrium ☐ Frozen

For Frozen Option, select NFZ value: ☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
Ref: [NASA RP1311 Part II \(Users Manual\), p. 18-19](#)

Select: ☒ Infinite Area Combustor ☐ Finite Area Combustor

For **Finite Area Combustor only**, select:

☒ Mass Flux/Chamber Area, kg/sec-sqm ☐ Contraction Ratio

Mass Flux or Contraction Ratio value:

Ref: [NASA RP1311 Part II \(Users Manual\), p.18-19](#)

Other Parameters

Name	Definition	Qty.	Value(s)
P	Pressure (atm)	3	100, 155, 210
OFP	o/f	8	1, 2, 3, 4, 5, 6, 7, 8
arrplot		3	gam, t, isp

What do you want to do upon clicking 'Submit'?

- ☒ Perform CEA Analysis. ☐ Tabulate results for insertion into a spreadsheet.
☐ Change Problem Type ☐ Select species for 'Omit/Only/Insert' options.

Submit input & Perform CEA Analysis.

Reset

ROCKET PROBLEM (RO)

Problem Type | Pressures | Fuel(s) | Oxidizer(s) | Oxid/Fuel | Exit Conditions | Final

Fuel(s): Pure H2

Oxid(s): Pure O2

Transfer CEA Results into a Spreadsheet

Enter up to 8 parameters, using either the Parameter Codes (see tables below) or names of chemical species (eg, H2, H2O, H).

IMPORTANT: Chemical names must exactly match those in the thermodynamic table. Do not use special characters, except for parentheses as used to indicate condensates, eg H2O(L).

*Ref: [NASA RP1311 Part II \(Users Manual\)](#), Section 2.5.4

Parameters:

-
-
-
-
-
-
-
-

Submit

Reset

Parameter Codes

All codes must be lowercase and match those in the tables below.

Thermodynamic Properties

Parameter	Code
Cross-section Area/Throat area	aeat
Density	rho
Enthalpy	h
Entropy	s
Gamma	gam
Gibbs Energy	g
Internal Energy	u
Molecular Wt. (1/n)	m
Molecular Wt. (M)	mw
Pressure	p
Sonic Velocity	son
Specific Heat	cp
Temperature	t

Rocket Problems Only

Parameter	Code
Area Ratio	ae
Coefficient of Thrust	cf
Ivac	ivac
Mach No	mach
Pe/Pe	pip
Specific Impulse	isp

For data from the Rocket Problem Frozen Tables, enter the appropriate parameter code, followed by the letters 'fz'.

Instructions

Numerical values use the same units as in the CEA Output File.

Importing .plt file Data into Excel

- From your plot screen, use your mouse to copy the columns you wish to import.
- 'Select' column A.
- Open a new Excel page and paste the data into box 'A1'. The data will now all be in that cell.
- The version of Excel you have dictates what you do next:
 - If you have a pick at the top called 'Data Select', click on it, and then 'Use Fixed width'
 - If there is no choice called 'Select', use 'Text to Columns', 'Fixed Width' and then 'Finish'
- You now have the data in the form you need for data workup or plotting.

NASA-GLENN CHEMICAL EQUILIBRIUM PROGRAM CEA2, FEBRUARY 5, 2004
BY BONNIE MCBRIDE AND SANFORD GORDON
REFS: NASA RP-1311, PART I, 1994 AND NASA RP-1311, PART II, 1996

CEA analysis performed on Thu 24-Oct-2024 21:38:33

Problem Type: "Rocket" (Infinite Area Combustor)

prob case=_____4070 ro equilibrium

Pressure (3 values):

p,atm= 100, 155, 210

Oxidizer/Fuel Wt. ratio (8 values):

o/f= 1, 2, 3, 4, 5, 6, 7, 8

You selected the following fuels and oxidizers:

reac

fuel H2 wt%=100.0000

oxid O2 wt%=100.0000

You selected these options for output:

short version of output

output short

Proportions of any products will be expressed as Mole Fractions.

Heat will be expressed as siunits

output siunits

Plot parameters:

output plot gam t isp

Input prepared by this

script:/var/www/sites/cearun.grc.nasa.gov/cgi-bin/CEARU

N/prepareInputFile.cgi

IMPORTANT: The following line is the end of your CEA input file!
end

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 1469.6 PSIA

CASE = _____

	REACTANT	WT FRACTION	ENERGY
TEMP		(SEE NOTE)	KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
0.000			
OXIDANT	O2	1.0000000	0.000
0.000			

O/F= 1.00000 %FUEL= 50.000000 R, EQ. RATIO= 7.936683
PHI, EQ. RATIO= 7.936683

	CHAMBER	THROAT
Pinf/P	1.0000	1.8638
P, BAR	101.33	54.365
T, K	1282.32	1093.65
RHO, KG/CU M	3.8316 0	2.4105 0
H, KJ/KG	0.00000	-1522.76
U, KJ/KG	-2644.46	-3778.15
G, KJ/KG	-46872.5	-41499.1
S, KJ/(KG) (K)	36.5530	36.5530

M, (1/n)	4.032	4.032
(dLV/dLP) t	-1.00000	-1.00000
(dLV/dLT) p	1.0000	1.0000
Cp, KJ/(KG) (K)	8.1963	7.9489
GAMMA s	1.3362	1.3503
SON VEL, M/SEC	1879.8	1745.1
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2408.7

CF	0.7245
Ivac, M/SEC	3037.5
Isp, M/SEC	1745.1

MOLE FRACTIONS

*H2	0.87400	0.87400
H2O	0.12600	0.12600

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 2277.9 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION	ENERGY
		(SEE NOTE)	KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
0.000			
OXIDANT	O2	1.0000000	0.000
0.000			

O/F= 1.00000 %FUEL= 50.000000 R, EQ. RATIO= 7.936683
PHI, EQ. RATIO= 7.936683

	CHAMBER	THROAT
Pinf/P	1.0000	1.8638
P, BAR	157.05	84.266
T, K	1282.32	1093.65
RHO, KG/CU M	5.9390 0	3.7362 0
H, KJ/KG	0.00010	-1522.76
U, KJ/KG	-2644.46	-3778.15
G, KJ/KG	-45713.6	-40510.6
S, KJ/(KG) (K)	35.6492	35.6492
M, (1/n)	4.032	4.032

(dLV/dLP) t	-1.00000	-1.00000
(dLV/dLT) p	1.0000	1.0000
Cp, KJ/(KG) (K)	8.1963	7.9489
GAMMA _s	1.3362	1.3503
SON VEL, M/SEC	1879.8	1745.1
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2408.7
CF	0.7245
Ivac, M/SEC	3037.5
Isp, M/SEC	1745.1

MOLE FRACTIONS

*H2	0.87400	0.87400
H2O	0.12600	0.12600

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 3086.2 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 1.00000 %FUEL= 50.000000 R, EQ. RATIO= 7.936683
PHI, EQ. RATIO= 7.936683

	CHAMBER	THROAT
P _{inf} /P	1.0000	1.8638
P, BAR	212.78	114.17
T, K	1282.32	1093.65
RHO, KG/CU M	8.0463 0	5.0619 0
H, KJ/KG	0.00004	-1522.76
U, KJ/KG	-2644.47	-3778.15
G, KJ/KG	-44910.5	-39825.7
S, KJ/(KG) (K)	35.0229	35.0229

M, (1/n)	4.032	4.032
(dLV/dLP) _t	-1.00000	-1.00000
(dLV/dLT) _p	1.0000	1.0000
C _p , KJ/(KG) (K)	8.1963	7.9489
GAMMA _s	1.3362	1.3503
SON VEL, M/SEC	1879.8	1745.1
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

A _e /A _t	1.0000
CSTAR, M/SEC	2408.7
CF	0.7245
I _{vac} , M/SEC	3037.5
I _{sp} , M/SEC	1745.1

MOLE FRACTIONS

*H ₂	0.87400	0.87400
H ₂ O	0.12600	0.12600

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

P_{in} = 1469.6 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION	ENERGY
		(SEE NOTE)	KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
0.000			
OXIDANT	O2	1.0000000	0.000
0.000			

O/F= 2.00000 %FUEL= 33.33333 R, EQ. RATIO= 3.968341
 PHI, EQ. RATIO= 3.968341

	CHAMBER	THROAT
Pinf/P	1.0000	1.8216
P, BAR	101.33	55.624
T, K	2073.91	1822.53
RHO, KG/CU M	3.5532 0	2.2199 0
H, KJ/KG	0.00000	-1604.81
U, KJ/KG	-2851.62	-4110.56
G, KJ/KG	-60159.3	-54472.2
S, KJ/(KG) (K)	29.0077	29.0077
M, (1/n)	6.047	6.047
(dLV/dLP) t	-1.00006	-1.00001
(dLV/dLT) p	1.0016	1.0004
Cp, KJ/(KG) (K)	6.5075	6.2737
GAMMA _s	1.2689	1.2809
SON VEL, M/SEC	1902.2	1791.5
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2547.8
CF	0.7032
Ivac, M/SEC	3190.2
Isp, M/SEC	1791.5

MOLE FRACTIONS

*H	0.00023	0.00005
*H2	0.74781	0.74796
H2O	0.25195	0.25199
*OH	0.00001	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 2277.9 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 2.00000 %FUEL= 33.33333 R, EQ. RATIO= 3.968341
PHI, EQ. RATIO= 3.968341

	CHAMBER	THROAT
Pinf/P	1.0000	1.8217
P, BAR	157.05	86.214
T, K	2074.18	1822.56
RHO, KG/CU M	5.5069 0	3.4406 0
H, KJ/KG	0.00000	-1604.99
U, KJ/KG	-2851.93	-4110.77
G, KJ/KG	-58917.3	-53375.1
S, KJ/(KG) (K)	28.4051	28.4051
M, (1/n)	6.047	6.048
(dLV/dLP) t	-1.00005	-1.00001
(dLV/dLT) p	1.0013	1.0003
Cp, KJ/(KG) (K)	6.4963	6.2706
GAMMA s	1.2693	1.2810
SON VEL, M/SEC	1902.6	1791.6
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2547.8
CF	0.7032
Ivac, M/SEC	3190.2
Isp, M/SEC	1791.6

MOLE FRACTIONS

*H	0.00018	0.00004
*H2	0.74785	0.74797
H2O	0.25196	0.25199
*OH	0.00001	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 3086.2 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 2.00000 %FUEL= 33.33333 R, EQ. RATIO= 3.968341
PHI, EQ. RATIO= 3.968341

	CHAMBER	THROAT
Pinf/P	1.0000	1.8217
P, BAR	212.78	116.80
T, K	2074.34	1822.58
RHO, KG/CU M	7.4605 0	4.6614 0
H, KJ/KG	0.00000	-1605.10

U, KJ/KG	-2852.11	-4110.89
G, KJ/KG	-58055.7	-52614.7
S, KJ/(KG) (K)	27.9876	27.9876

M, (1/n)	6.047	6.048
(dLV/dLP) t	-1.00004	-1.00001
(dLV/dLT) p	1.0011	1.0003
Cp, KJ/(KG) (K)	6.4899	6.2688
GAMMA _s	1.2695	1.2811
SON VEL, M/SEC	1902.8	1791.7
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2547.8
CF	0.7032
Ivac, M/SEC	3190.3
Isp, M/SEC	1791.7

MOLE FRACTIONS

*H	0.00016	0.00003
*H2	0.74787	0.74798
H2O	0.25197	0.25199
*OH	0.00001	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 1469.6 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			

FUEL	H2	1.0000000	0.000
0.000			
OXIDANT	O2	1.0000000	0.000
0.000			

O/F= 3.00000 %FUEL= 25.000000 R, EQ. RATIO= 2.645561
 PHI, EQ. RATIO= 2.645561

	CHAMBER	THROAT
Pinf/P	1.0000	1.7920
P, BAR	101.33	56.545
T, K	2699.13	2427.93
RHO, KG/CU M	3.6312 0	2.2562 0
H, KJ/KG	0.00000	-1543.91
U, KJ/KG	-2790.39	-4050.11
G, KJ/KG	-66154.4	-61051.3
S, KJ/(KG) (K)	24.5095	24.5095
M, (1/n)	8.043	8.055
(dLV/dLP) t	-1.00130	-1.00054
(dLV/dLT) p	1.0275	1.0125
Cp, KJ/(KG) (K)	6.0224	5.6020
GAMMA s	1.2194	1.2321
SON VEL, M/SEC	1844.6	1757.2
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2555.7
CF	0.6876
Ivac, M/SEC	3183.4
Isp, M/SEC	1757.2

MOLE FRACTIONS

*H	0.00443	0.00190
*H2	0.61856	0.62051
H2O	0.37625	0.37734
*OH	0.00075	0.00024

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 2277.9 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
0.000			
OXIDANT	O2	1.0000000	0.000
0.000			

O/F= 3.00000 %FUEL= 25.000000 R, EQ. RATIO= 2.645561
 PHI, EQ. RATIO= 2.645561

	CHAMBER	THROAT
Pinf/P	1.0000	1.7929
P, BAR	157.05	87.597
T, K	2704.24	2429.63
RHO, KG/CU M	5.6204 0	3.4935 0
H, KJ/KG	0.00000	-1546.80
U, KJ/KG	-2794.37	-4054.24
G, KJ/KG	-65054.8	-59995.4
S, KJ/(KG) (K)	24.0565	24.0565
M, (1/n)	8.046	8.057
(dLV/dLP) t	-1.00107	-1.00044
(dLV/dLT) p	1.0225	1.0101
Cp, KJ/(KG) (K)	5.9128	5.5449
GAMMA _s	1.2220	1.2338
SON VEL, M/SEC	1847.9	1758.9
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2556.0

CF	0.6881
Ivac, M/SEC	3184.5
Isp, M/SEC	1758.9

MOLE FRACTIONS

*H	0.00363	0.00154
*H2	0.61918	0.62080
H2O	0.37656	0.37746
*OH	0.00062	0.00020

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 3086.2 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 3.00000 %FUEL= 25.000000 R, EQ. RATIO= 2.645561
PHI, EQ. RATIO= 2.645561

	CHAMBER	THROAT
Pinf/P	1.0000	1.7935
P, BAR	212.78	118.64
T, K	2707.27	2430.62
RHO, KG/CU M	7.6083 0	4.7302 0
H, KJ/KG	0.00000	-1548.50
U, KJ/KG	-2796.72	-4056.66
G, KJ/KG	-64278.2	-59258.2
S, KJ/(KG) (K)	23.7428	23.7428

M, (1/n)	8.049	8.057
(dLV/dLP) t	-1.00093	-1.00038
(dLV/dLT) p	1.0195	1.0087
Cp, KJ/(KG) (K)	5.8482	5.5117
GAMMA _s	1.2235	1.2348
SON VEL, M/SEC	1849.8	1759.8
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2556.1
CF	0.6885
Ivac, M/SEC	3185.1
Isp, M/SEC	1759.8

MOLE FRACTIONS

*H	0.00316	0.00133
*H ₂	0.61955	0.62096
H ₂ O	0.37675	0.37754
*OH	0.00054	0.00017

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 1469.6 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H ₂	1.0000000	0.000
OXIDANT	O ₂	1.0000000	0.000

O/F= 4.00000 %FUEL= 20.000000 R, EQ. RATIO= 1.984171
 PHI, EQ. RATIO= 1.984171

	CHAMBER	THROAT
Pinf/P	1.0000	1.7654
P, BAR	101.33	57.394
T, K	3156.16	2903.95
RHO, KG/CU M	3.8438 0	2.3785 0
H, KJ/KG	0.00000	-1434.44
U, KJ/KG	-2636.03	-3847.46
G, KJ/KG	-68009.3	-64009.1
S, KJ/(KG) (K)	21.5481	21.5481
M, (1/n)	9.955	10.006
(dLV/dLP) t	-1.00626	-1.00367
(dLV/dLT) p	1.1176	1.0741
Cp, KJ/(KG) (K)	6.5719	5.8975
GAMMA _s	1.1799	1.1889
SON VEL, M/SEC	1763.6	1693.8
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2515.1
CF	0.6734
Ivac, M/SEC	3118.4
Isp, M/SEC	1693.8

MOLE FRACTIONS

*H	0.01727	0.01076
*H2	0.48502	0.48893
H2O	0.49051	0.49659
*O	0.00014	0.00005
*OH	0.00701	0.00365
*O2	0.00005	0.00002

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 2277.9 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 4.00000 %FUEL= 20.000000 R, EQ. RATIO= 1.984171
 PHI, EQ. RATIO= 1.984171

	CHAMBER	THROAT
Pinf/P	1.0000	1.7677
P, BAR	157.05	88.846
T, K	3175.66	2914.88
RHO, KG/CU M	5.9324 0	3.6726 0
H, KJ/KG	0.00000	-1442.62
U, KJ/KG	-2647.38	-3861.79
G, KJ/KG	-67268.2	-63186.9
S, KJ/(KG) (K)	21.1824	21.1824
M, (1/n)	9.974	10.018
(dLV/dLP) t	-1.00531	-1.00306
(dLV/dLT) p	1.0992	1.0616
Cp, KJ/(KG) (K)	6.2744	5.6825
GAMMA _s	1.1837	1.1927
SON VEL, M/SEC	1770.3	1698.6
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2517.6
CF	0.6747
Ivac, M/SEC	3122.8
Isp, M/SEC	1698.6

MOLE FRACTIONS

*H	0.01466	0.00898
*H2	0.48668	0.49011
H2O	0.49248	0.49780
*O	0.00010	0.00003
*OH	0.00604	0.00307
*O2	0.00004	0.00001

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 3086.2 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 4.00000 %FUEL= 20.000000 R, EQ. RATIO= 1.984171
PHI, EQ. RATIO= 1.984171

	CHAMBER	THROAT
Pinf/P	1.0000	1.7691
P, BAR	212.78	120.28
T, K	3187.83	2921.52
RHO, KG/CU M	8.0161 0	4.9642 0
H, KJ/KG	0.00000	-1447.66
U, KJ/KG	-2654.43	-3870.55
G, KJ/KG	-66719.4	-62593.2
S, KJ/(KG) (K)	20.9294	20.9294

M, (1/n)	9.985	10.026
(dLV/dLP) _t	-1.00472	-1.00269
(dLV/dLT) _p	1.0879	1.0540
C _p , KJ/(KG) (K)	6.0927	5.5535
GAMMA _s	1.1863	1.1950
SON VEL, M/SEC	1774.5	1701.6
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

A _e /A _t	1.0000
CSTAR, M/SEC	2519.1
CF	0.6755
I _{vac} , M/SEC	3125.5
I _{sp} , M/SEC	1701.6

MOLE FRACTIONS

*H	0.01304	0.00789
*H ₂	0.48771	0.49082
H ₂ O	0.49372	0.49854
*O	0.00008	0.00002
*OH	0.00541	0.00271
*O ₂	0.00003	0.00001

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

P_{in} = 1469.6 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION	ENERGY
		(SEE NOTE)	KJ/KG-MOL
K			
FUEL	H ₂	1.0000000	0.000
0.000			

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 2277.9 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 5.00000 %FUEL= 16.666667 R, EQ. RATIO= 1.587337
PHI, EQ. RATIO= 1.587337

	CHAMBER	THROAT
Pinf/P	1.0000	1.7485
P, BAR	157.05	89.821
T, K	3497.40	3264.66
RHO, KG/CU M	6.3564 0	3.9263 0
H, KJ/KG	0.00000	-1328.94
U, KJ/KG	-2470.79	-3616.59
G, KJ/KG	-66937.6	-63812.0
S, KJ/(KG) (K)	19.1392	19.1392
M, (1/n)	11.769	11.865
(dLV/dLP) t	-1.01416	-1.00988
(dLV/dLT) p	1.2477	1.1842
Cp, KJ/(KG) (K)	7.2924	6.5865
GAMMA s	1.1583	1.1618
SON VEL, M/SEC	1691.7	1630.3
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At 1.0000

CSTAR, M/SEC	2453.5
CF	0.6645
Ivac, M/SEC	3033.5
Isp, M/SEC	1630.3

MOLE FRACTIONS

*H	0.02813	0.02118
H02	0.00001	0.00000
*H2	0.35949	0.36113
H20	0.58815	0.60168
H202	0.00001	0.00000
*O	0.00096	0.00049
*OH	0.02266	0.01520
*O2	0.00060	0.00032

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 3086.2 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION	ENERGY
K		(SEE NOTE)	KJ/KG-MOL
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 5.00000 %FUEL= 16.666667 R, EQ. RATIO= 1.587337
PHI, EQ. RATIO= 1.587337

	CHAMBER	THROAT
Pinf/P	1.0000	1.7501
P, BAR	212.78	121.58
T, K	3522.24	3282.22

RHO, KG/CU M	8.5720	0 5.2962	0
H, KJ/KG	0.00000	-1336.58	
U, KJ/KG	-2482.28	-3632.17	
G, KJ/KG	-66658.2	-63452.4	
S, KJ/(KG) (K)	18.9250	18.9250	

M, (1/n)	11.798	11.888
(dLV/dLP) t	-1.01287	-1.00888
(dLV/dLT) p	1.2237	1.1647
Cp, KJ/(KG) (K)	6.9743	6.3198
GAMMA _s	1.1607	1.1645
SON VEL, M/SEC	1697.4	1635.0
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2457.3
CF	0.6654
Ivac, M/SEC	3039.0
Isp, M/SEC	1635.0

MOLE FRACTIONS

*H	0.02559	0.01907
H ₂	0.00001	0.00000
*H ₂	0.36044	0.36200
H ₂ O	0.59169	0.60439
H ₂ O ₂	0.00001	0.00000
*O	0.00080	0.00040
*OH	0.02095	0.01387
*O ₂	0.00051	0.00026

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 1469.6 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION	ENERGY
		(SEE NOTE)	KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
0.000			
OXIDANT	O2	1.0000000	0.000
0.000			

O/F= 6.00000 %FUEL= 14.285714 R, EQ. RATIO= 1.322780
 PHI, EQ. RATIO= 1.322780

	CHAMBER	THROAT
Pinf/P	1.0000	1.7346
P, BAR	101.33	58.415
T, K	3637.50	3439.83
RHO, KG/CU M	4.4628 0	2.7528 0
H, KJ/KG	0.00000	-1209.08
U, KJ/KG	-2270.45	-3331.08
G, KJ/KG	-65040.1	-62714.7
S, KJ/(KG) (K)	17.8804	17.8804
M, (1/n)	13.321	13.478
(dLV/dLP) t	-1.03046	-1.02451
(dLV/dLT) p	1.5191	1.4417
Cp, KJ/(KG) (K)	9.3701	8.7241
GAMMA s	1.1406	1.1396
SON VEL, M/SEC	1609.2	1555.0
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2367.0
CF	0.6570
Ivac, M/SEC	2919.6
Isp, M/SEC	1555.0

MOLE FRACTIONS

*H	0.03974	0.03353
HO2	0.00005	0.00003
*H2	0.25120	0.24775

H2O	0.64585	0.66900
H2O2	0.00002	0.00001
*O	0.00459	0.00318
*OH	0.05404	0.04320
*O2	0.00451	0.00331

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 2277.9 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
0.000			
OXIDANT	O2	1.0000000	0.000
0.000			

O/F= 6.00000 %FUEL= 14.285714 R, EQ. RATIO= 1.322780
PHI, EQ. RATIO= 1.322780

	CHAMBER	THROAT
Pinf/P	1.0000	1.7364
P, BAR	157.05	90.446
T, K	3691.88	3484.36
RHO, KG/CU M	6.8524 0	4.2280 0
H, KJ/KG	0.00000	-1222.12
U, KJ/KG	-2291.94	-3361.32
G, KJ/KG	-65005.3	-62573.4
S, KJ/(KG) (K)	17.6076	17.6076
M, (1/n)	13.393	13.543
(dLV/dLP) t	-1.02757	-1.02189
(dLV/dLT) p	1.4641	1.3906
Cp, KJ/(KG) (K)	8.7032	8.0935

GAMMA _s	1.1433	1.1426
SON VEL, M/SEC	1618.8	1563.4
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

A _e /A _t	1.0000
CSTAR, M/SEC	2376.0
CF	0.6580
I _{vac} , M/SEC	2931.7
I _{sp} , M/SEC	1563.4

MOLE FRACTIONS

*H	0.03569	0.02985
H ₂ O	0.00005	0.00003
*H ₂	0.25071	0.24741
H ₂ O	0.65519	0.67738
H ₂ O ₂	0.00002	0.00001
*O	0.00386	0.00261
*OH	0.05063	0.03995
*O ₂	0.00384	0.00275

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

P_{in} = 3086.2 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H ₂	1.0000000	0.000
OXIDANT	O ₂	1.0000000	0.000

O/F= 6.00000 %FUEL= 14.285714 R, EQ. RATIO= 1.322780
 PHI, EQ. RATIO= 1.322780

	CHAMBER	THROAT
Pinf/P	1.0000	1.7378
P, BAR	212.78	122.45
T, K	3728.56	3514.02
RHO, KG/CU M	9.2264 0	5.6938 0
H, KJ/KG	0.00000	-1230.87
U, KJ/KG	-2306.25	-3381.40
G, KJ/KG	-64949.4	-62443.1
S, KJ/(KG) (K)	17.4194	17.4194
M, (1/n)	13.442	13.586
(dLV/dLP) t	-1.02561	-1.02015
(dLV/dLT) p	1.4277	1.3571
Cp, KJ/(KG) (K)	8.2733	7.6898
GAMMA _s	1.1452	1.1447
SON VEL, M/SEC	1625.2	1569.0
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2381.8
CF	0.6587
Ivac, M/SEC	2939.6
Isp, M/SEC	1569.0

MOLE FRACTIONS

*H	0.03300	0.02742
H ₂ O	0.00006	0.00003
*H ₂	0.25035	0.24716
H ₂ O	0.66162	0.68307
H ₂ O ₂	0.00002	0.00001
*O	0.00339	0.00226
*OH	0.04815	0.03764
*O ₂	0.00341	0.00240

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 1469.6 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 7.00000 %FUEL= 12.500000 R, EQ. RATIO= 1.133812
 PHI, EQ. RATIO= 1.133812

	CHAMBER	THROAT
Pinf/P	1.0000	1.7292
P, BAR	101.33	58.595
T, K	3718.05	3533.55
RHO, KG/CU M	4.8252 0	2.9752 0
H, KJ/KG	0.00000	-1113.85
U, KJ/KG	-2099.90	-3083.32
G, KJ/KG	-61943.6	-59983.7
S, KJ/(KG) (K)	16.6603	16.6603
M, (1/n)	14.722	14.918
(dLV/dLP)t	-1.04337	-1.03806
(dLV/dLT)p	1.7246	1.6700
Cp, KJ/(KG) (K)	10.3882	10.0950
GAMMA _s	1.1342	1.1311
SON VEL, M/SEC	1543.3	1492.6
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2281.8
CF	0.6541
Ivac, M/SEC	2812.1

Isp, M/SEC 1492.5

MOLE FRACTIONS

*H	0.03877	0.03373
H2	0.00014	0.00009
*H2	0.17165	0.16399
H2O	0.67954	0.70756
H2O2	0.00004	0.00002
*O	0.01018	0.00812
*OH	0.08434	0.07303
*O2	0.01535	0.01345

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 2277.9 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
0.000			
OXIDANT	O2	1.0000000	0.000
0.000			

O/F= 7.00000 %FUEL= 12.500000 R, EQ. RATIO= 1.133812
PHI, EQ. RATIO= 1.133812

	CHAMBER	THROAT
Pinf/P	1.0000	1.7306
P, BAR	157.05	90.752
T, K	3782.07	3589.13
RHO, KG/CU M	7.4004 0	4.5640 0
H, KJ/KG	0.00000	-1126.74
U, KJ/KG	-2122.24	-3115.16

G, KJ/KG	-62077.1	-60037.0
S, KJ/(KG) (K)	16.4135	16.4135
M, (1/n)	14.817	15.008
(dLV/dLP) _t	-1.04043	-1.03527
(dLV/dLT) _p	1.6660	1.6130
C _p , KJ/(KG) (K)	9.7107	9.4276
GAMMA _s	1.1363	1.1333
SON VEL, M/SEC	1552.9	1501.2
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

A _e /A _t	1.0000
CSTAR, M/SEC	2292.3
CF	0.6549
I _{vac} , M/SEC	2825.8
I _{sp} , M/SEC	1501.2

MOLE FRACTIONS

*H	0.03505	0.03035
H ₂ O	0.00016	0.00010
*H ₂	0.16879	0.16114
H ₂ O	0.69172	0.71924
H ₂ O ₂	0.00005	0.00003
*O	0.00899	0.00709
*OH	0.08127	0.06993
*O ₂	0.01398	0.01212

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

P_{in} = 3086.2 PSIA
CASE = _____

TEMP	REACTANT	WT FRACTION	ENERGY
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		(SEE NOTE)	KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
0.000			
OXIDANT	O2	1.0000000	0.000
0.000			

O/F= 7.00000 %FUEL= 12.500000 R, EQ. RATIO= 1.133812
 PHI, EQ. RATIO= 1.133812

	CHAMBER	THROAT
Pinf/P	1.0000	1.7315
P, BAR	212.78	122.89
T, K	3826.11	3627.17
RHO, KG/CU M	9.9555 0	6.1407 0
H, KJ/KG	0.00000	-1135.50
U, KJ/KG	-2137.34	-3136.69
G, KJ/KG	-62149.5	-60053.4
S, KJ/(KG) (K)	16.2435	16.2435
M, (1/n)	14.884	15.070
(dLV/dLP) t	-1.03840	-1.03335
(dLV/dLT) p	1.6266	1.5747
Cp, KJ/(KG) (K)	9.2689	8.9923
GAMMA _s	1.1377	1.1348
SON VEL, M/SEC	1559.4	1507.0
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2299.4
CF	0.6554
Ivac, M/SEC	2834.9
Isp, M/SEC	1507.0

MOLE FRACTIONS

*H	0.03257	0.02810
H02	0.00018	0.00011
*H2	0.16671	0.15909
H20	0.70032	0.72743
H202	0.00005	0.00003
*O	0.00819	0.00642

*OH	0.07893	0.06761
*O2	0.01304	0.01122

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 1469.6 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 8.00000 %FUEL= 11.111111 R, EQ. RATIO= 0.992085
 PHI, EQ. RATIO= 0.992085

	CHAMBER	THROAT
Pinf/P	1.0000	1.7279
P, BAR	101.33	58.641
T, K	3731.74	3550.68
RHO, KG/CU M	5.2031 0	3.2077 0
H, KJ/KG	0.00000	-1031.95
U, KJ/KG	-1947.39	-2860.04
G, KJ/KG	-58525.4	-56717.7
S, KJ/(KG) (K)	15.6831	15.6831
M, (1/n)	15.933	16.149
(dLV/dLP) t	-1.04701	-1.04221
(dLV/dLT) p	1.7825	1.7393
Cp, KJ/(KG) (K)	10.1108	9.9545
GAMMA s	1.1325	1.1290
SON VEL, M/SEC	1485.0	1436.6
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1. 0000
CSTAR, M/SEC	2198. 7
CF	0. 6534
Ivac, M/SEC	2709. 1
Isp, M/SEC	1436. 6

MOLE FRACTIONS

*H	0. 03301	0. 02852
H02	0. 00028	0. 00019
*H2	0. 11781	0. 10878
H20	0. 69020	0. 71920
H202	0. 00006	0. 00004
*0	0. 01601	0. 01351
*OH	0. 10695	0. 09548
*02	0. 03568	0. 03428

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 2277. 9 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION	ENERGY
		(SEE NOTE)	KJ/KG-MOL
K			
FUEL	H2	1. 0000000	0. 000
0. 000			
OXIDANT	02	1. 0000000	0. 000
0. 000			

O/F= 8. 00000 %FUEL= 11. 111111 R, EQ. RATIO= 0. 992085
 PHI, EQ. RATIO= 0. 992085

	CHAMBER	THROAT
P _{inf} /P	1.0000	1.7291
P, BAR	157.05	90.830
T, K	3797.78	3608.75
RHO, KG/CU M	7.9785 0	4.9198 0
H, KJ/KG	0.00000	-1043.98
U, KJ/KG	-1968.47	-2890.19
G, KJ/KG	-58695.4	-56817.9
S, KJ/(KG) (K)	15.4552	15.4552

M, (1/n)	16.041	16.252
(dLV/dLP) _t	-1.04424	-1.03960
(dLV/dLT) _p	1.7254	1.6842
C _p , KJ/(KG) (K)	9.4842	9.3396
GAMMA _s	1.1344	1.1309
SON VEL, M/SEC	1494.3	1445.0
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

A _e /A _t	1.0000
CSTAR, M/SEC	2209.2
CF	0.6541
I _{vac} , M/SEC	2722.7
I _{sp} , M/SEC	1445.0

MOLE FRACTIONS

*H	0.02964	0.02548
H ₂ O	0.00033	0.00023
*H ₂	0.11363	0.10453
H ₂ O	0.70321	0.73185
H ₂ O ₂	0.00007	0.00005
*O	0.01451	0.01220
*OH	0.10453	0.09301
*O ₂	0.03408	0.03267

* THERMODYNAMIC PROPERTIES FITTED TO 20000. K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 3086.2 PSIA

CASE = _____

TEMP	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL
K			
FUEL	H2	1.0000000	0.000
OXIDANT	O2	1.0000000	0.000

O/F= 8.00000 %FUEL= 11.11111 R, EQ. RATIO= 0.992085
 PHI, EQ. RATIO= 0.992085

	CHAMBER	THROAT
Pinf/P	1.0000	1.7299
P, BAR	212.78	123.00
T, K	3843.45	3648.77
RHO, KG/CU M	1.0731	6.6183
H, KJ/KG	0.00000	-1052.16
U, KJ/KG	-1982.79	-2910.67
G, KJ/KG	-58797.6	-56871.6
S, KJ/(KG) (K)	15.2982	15.2982
M, (1/n)	16.117	16.324
(dLV/dLP) t	-1.04234	-1.03782
(dLV/dLT) p	1.6872	1.6474
Cp, KJ/(KG) (K)	9.0764	8.9393
GAMMA _s	1.1357	1.1323
SON VEL, M/SEC	1500.6	1450.6
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2216.3
CF	0.6545
Ivac, M/SEC	2731.8
Isp, M/SEC	1450.6

MOLE FRACTIONS

*H	0.02741	0.02347
H02	0.00037	0.00025
*H2	0.11059	0.10147
H20	0.71244	0.74078
H202	0.00009	0.00006
*O	0.01351	0.01131
*OH	0.10265	0.09112
*O2	0.03295	0.03154

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS